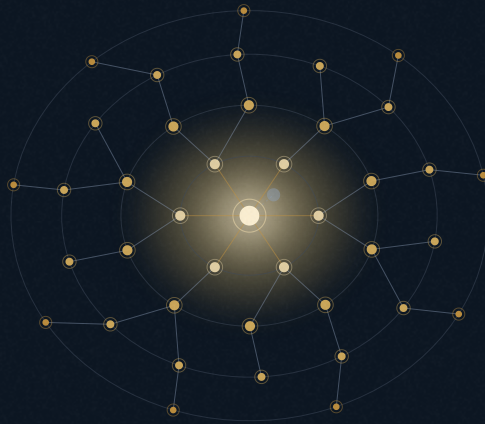




KRONOS FUSION ENERGY



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Causal Intervention Models for Regulator-Facing Plasma Control: Do-Calculus Prediction and Per-Decision Explainability in a Compact Spherical-Tokamak Breeder

A controller that holds a nuclear plasma must answer, on every cycle, two questions a regulator will ask: what will this action do, and why did you take it. Correlational machine...

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THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TWO EDITIONS

This volume is issued in two editions. The **public edition** (light) carries the physics and the design only — no dollar, price, or per-kilo-gram figure appears anywhere in it. The **internal edition** (dark) adds the economics, the funding architecture, and the defense-commercial analysis, and is marked *Confidential* accordingly; it is not for public distribution. Where the commercial case rests on a modelled assumption rather than a demonstrated one, it is labelled as such and its downside is shown alongside it, on the same terms as the physics.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

Causal Intervention Models for Regulator-Facing Plasma Control: Do-Calculus Prediction and Per-Decision Explainability in a Compact Spherical-Tokamak Breeder

A controller that holds a nuclear plasma must answer, on every cycle, two questions a regulator will ask: *what will this action do*, and *why did you take it*. Correlational machine learning answers neither; both are causal questions, and a control action is not an observation but an *intervention*. We formalise the Kronos digital twin of the Hyperion breeder (design point $I_p = 9.66\text{ MA}$, $Q = 3.076$, $P_{\text{fus}} = 85.04\text{ MW}$, negative triangularity $\delta = -0.30$, $B_0 = 8\text{ T}$, elongation $\kappa = 2.0$, major radius $R_0 = 1.20\text{ m}$, aspect ratio $A = 2.5$, centrepost peak field 16.84 T) as a differentiable structural causal model (SCM): its nodes are physical states, its edges are the physics couplings, and its mechanisms are physics operators, so a control action is an executable do-calculus intervention $\text{do}(\mathbf{u})$ whose interventional distribution the twin computes directly, and every decision carries a gradient-based attribution to the states that drove it. We write the SCM, the truncated-factorisation identity for $P(\mathbf{y} \mid \text{do}(\mathbf{u}))$, the control-barrier quadratic program that certifies the intervention, and the integrated-gradient and Sobol attributions, as explicit equations, and we discretise and solve each on the NVIDIA GPU high-performance-computing (HPC) campaign. Three subsystems are reduced to practice against pre-registered acceptance tests. A certified intervention on the normalised pressure — $\text{do}(\mathbf{u})$ with $\mathbf{u} \in [1.061, 1.250]$ — caps β_N at the safety barrier $\beta_N^{\text{max}} = 3.500$ with 0 of 100 boundary crossings and a certified minimum barrier value $h_{\text{min}} = +0.000$ (forward invariance), against a maximum of 4.200 and 50/100 crossings with no intervention. A physics-informed equilibrium operator reconstructs the poloidal flux to an RMS error of 0.139 % against a free-boundary Grad-Shafranov solver (acceptance < 1%; Grad-Shafranov operator residual 0.32%) at a 2877× speedup, and a graph-network state estimator reaches an area under the ROC curve of 0.980, above its single-channel baselines. A variance-based (Sobol) decomposition over a 40 000-sample Monte-Carlo ensemble assigns 88.5 % of the breeder- Q variance to the confinement factor H_{98} (first-order index $S_1 = 0.885$, total $S_T = 0.905$), on a calibrated ensemble in which the anchor $Q = 3.076$ sits inside the [1.34, 8.51] interquantile band and $H_{98} = 1.007 \pm 0.030$. The result is a controller whose interventions are certified and whose decisions are attributable — a working core for auditable autonomy — with the runtime per-decision explainer built to this same demonstrated pattern on a staged schedule.

Open-access deposit (code & data, CC BY 4.0): DOI [10.5281/zenodo.22645825](https://doi.org/10.5281/zenodo.22645825) · Zenodo community *kronos_fusion_energy*. Explore it live: [interactive 3-D model](#) · [physics validator](#).

Keywords: causal inference do-calculus explainability control-barrier function differentiable digital twin global sensitivity spherical-tokamak breeder

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

Introduction

AUTONOMY AND THE BURDEN OF PROOF

A controller that holds **9.66 MA** of plasma current in a compact volume behind a centrepost magnet at a peak field of **16.84 T** must do more than act well. In a nuclear system every actuation carries a safety consequence, the characteristic times of vertical instability and thermal transients are short, and the operator — and, downstream, the regulator — will require the machine to answer two questions on the record. The first is predictive: *what does a candidate action do* before it is taken. The second is explanatory: *why was the action that was taken the right one*. These are not questions of accuracy or confidence; they are questions of *cause*. An association between a diagnostic and an outcome, however tight, does not tell a controller what happens when it *intervenes*, and a saliency map that merely correlates with a decision does not justify it. The distinction is the one Pearl draws between seeing and doing^[1,2]: the observational law $P(y | u)$ and the interventional law $P(y | \text{do}(u))$ are different objects, and acting on the former where the latter is required is exactly how a confident controller makes a systematic mistake^[5].

PRIOR ART

Feedback control of tokamak equilibria and profiles is mature. Magnetic diagnostics feed real-time equilibrium reconstruction^[12] and shape controllers^[13]; physics-based profile simulators run inside the control cycle^[14]; and learned controllers have entered the loop, from a deep reinforcement-learning agent that commands the TCV coil voltages directly from magnetics^[15] to deep networks that forecast disruptions across machines^[16]. In parallel, the predictive digital twin has become a rigorous object^[17], made fast by neural operators^[18,19] and physics-informed training^[20], and made honest by ensemble assimilation^[27,28] and reduced-order modelling^[22]. What none of these lines supplies on its own is the causal and explanatory layer a regulator needs: control barrier functions give a certificate of *safety*^[6,7] but not of *why*; attribution methods give a *why*^[8,9] but no guarantee of physical faithfulness; and global-sensitivity analysis names the *governing* variable^[10,11] but at the population level, not per decision. Causal inference is the framework that ties them together, and causal representation learning is only now reaching control-relevant maturity^[4,3].

CONTRIBUTION

This paper makes the causal structure of the Kronos twin explicit and shows that it answers both questions by construction. Our contributions are: (i) a formal statement of the twin as a differentiable structural causal model, with the do-operator realised as an executable submodel and the interventional distribution written as a truncated factorisation (§2); (ii) a certified-intervention law — the control-barrier quadratic program that projects any proposed action onto the largest safe action and renders the operating set forward-invariant (§2.4); (iii) a per-decision explainability construction in which the differentiability of the twin’s mechanisms yields gradient, integrated-gradient and Shapley attributions with a completeness guarantee (§2.5); (iv) a population-level causal attribution by Sobol variance decomposition on a calibrated ensemble (§2.6); (v) a GPU-HPC methodology that names the codes, the discretisations, the solvers and the convergence checks that produced every number (§3); and (vi) a demonstration to pre-registered acceptance on three reduced-to-practice subsystems (§4). Every quantitative claim traces to a frozen anchor in the Kronos Physics De-Risking Register^[31]; the serial identifiers (e.g. KX-L1-A13) are given inline so each number is auditable. The certified-clamp, differentiable-surrogate and maturity-gated elements build directly on the companion control-stack results: the control-barrier safety filter^[33] ([official page](#)), the differentiable neural-operator flux surrogate^[34] ([official page](#)), the sub-millisecond neural equilibrium reconstructor^[35] ([official page](#)), and the four-coupled-digital-twin architecture^[36] ([official page](#)) inside which this causal layer sits.

§ 2

Governing equations and the formal problem

THE TWIN AS A STRUCTURAL CAUSAL MODEL

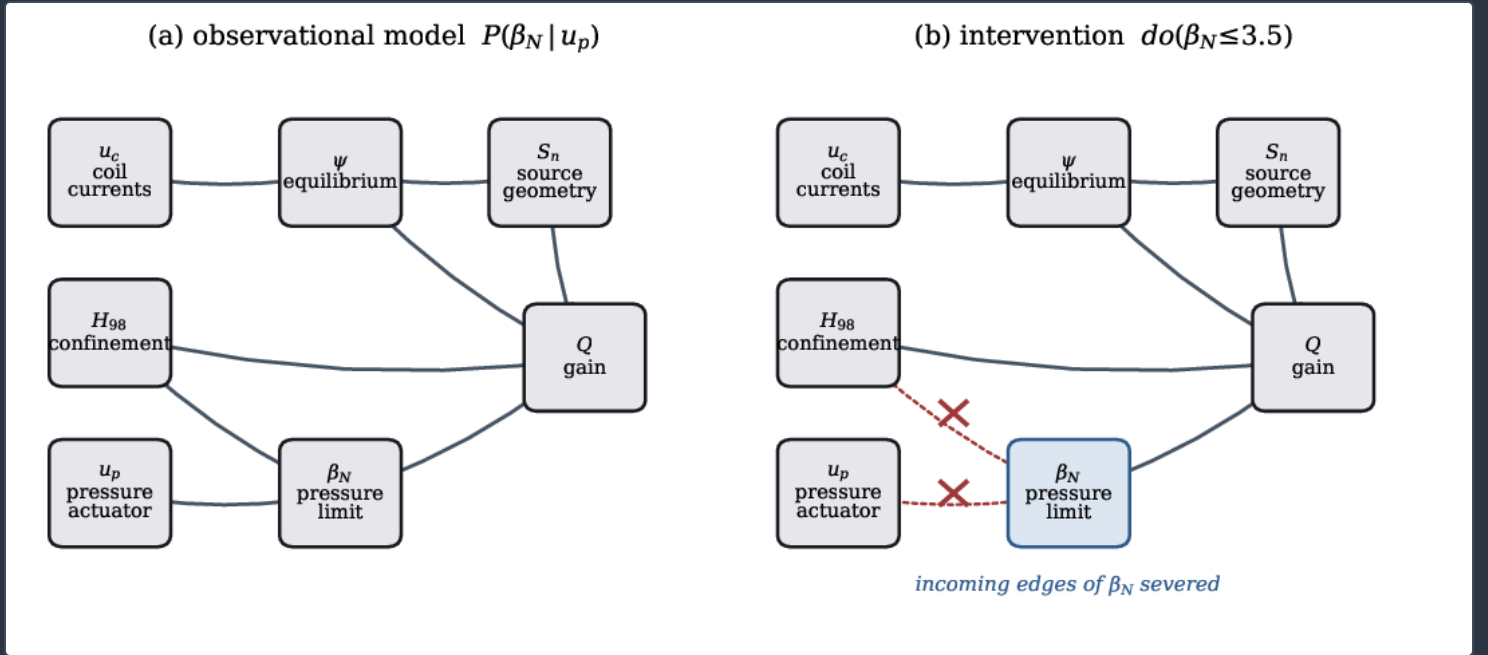
A structural causal model (SCM) is a tuple $\mathcal{M} = \langle U, V, \mathcal{F}, P(U) \rangle$ of exogenous variables U , endogenous variables $V = \{V_1, \dots, V_d\}$, structural mechanisms $\mathcal{F} = \{f_1, \dots, f_d\}$, and a distribution over the exogenous noise ^[2,3]. Each endogenous variable is generated from its parents and its own noise,

$$V_i = f_i(\text{pa}(V_i), U_i), \quad i = 1, \dots, d,$$

and the induced directed graph $\mathcal{G}$ is acyclic. The Kronos twin is exactly such a model. Its endogenous nodes are the physical states of the four-twin architecture ^[36] — the coil currents u_c , the poloidal-flux equilibrium ψ , the 14 MeV source geometry S_n , the confinement quality H_{98} , the normalised pressure β_N , and the scientific gain Q (figure 1) — and its edges are the physics couplings between them: the coil currents set the boundary field that fixes the equilibrium; the equilibrium sets the neutronics source geometry; the pressure actuator and confinement set β_N ; and the equilibrium, source and pressure set Q . Crucially, the mechanisms f_i are *not* statistical fits but physics operators — the Grad–Shafranov equation for ψ , the transport map for H_{98} , the power balance for Q — so the graph carries mechanism, not merely correlation, and the interventions defined on it are physically meaningful. Because the joint factorises along $\mathcal{G}$,

$$P(v_1, \dots, v_d) = \prod_{i=1}^d P(v_i \mid \text{pa}(v_i)),$$

the model is a compact, auditable encoding of the plant's dependence structure.



(Schematic.) The twin as a structural causal model, and the do-operator. **(a)** The observational graph: coil currents u_c , equilibrium ψ , source geometry S_n , confinement H_{98} , pressure actuator u_p , normalised pressure β_N and gain Q , with edges carrying the physics couplings. **(b)** The intervention $do(\beta_N \leq 3.5)$ used by the safety clamp severs the incoming edges of β_N (equation 3) and sets it, so the twin predicts the effect of the action rather than the correlation that co-occurs. Structure only; no data values are shown.

THE DO-OPERATOR AND THE INTERVENTIONAL DISTRIBUTION

A control action is an intervention. The atomic intervention $\text{do}(X=x)$ replaces the mechanism f_X of the intervened node with the constant assignment $X=x$, leaving every other mechanism unchanged; this produces the *mutilated* submodel $\mathcal{M}_x$ whose graph $\mathcal{G}_{\bar{X}}$ deletes the edges into X (figure 1b),

$$\text{do}(X=x) : f_X \mapsto (X=x), \quad \mathcal{G} \mapsto \mathcal{G}_{\bar{X}}.$$

The interventional distribution then follows from the *truncated factorisation* — the same product as equation (2) with the factor for the intervened node removed and its value fixed ^[2],

$$P(v_1, \dots, v_d \mid \text{do}(X=x)) = \prod_{i: V_i \neq X} P(v_i \mid \text{pa}(v_i)) \Big|_{X=x},$$

and is generically *not* equal to the conditional $P(v \mid X=x)$ obtained by observation, because conditioning propagates information backward through the parents of X while intervening does not. For a controller this is the whole point: $\text{do}(u)$ is the twin’s prediction of the plant *under the action*, and

$$P(y \mid \text{do}(u)) \neq P(y \mid u)$$

in general. When a back-door set Z blocks every spurious path from the actuator to the outcome, the interventional law is identified from observational data by the adjustment formula $P(y \mid \text{do}(u)) = \sum_z P(y \mid u, z) P(z)$; in the Kronos twin, however, the mechanisms are known physics operators, so equation (4) is evaluated *directly* by executing the mutilated submodel, and identification from confounded data is not required.

COUNTERFACTUALS ON THE SHADOW BRANCH

A counterfactual is the three-step abduction–action–prediction computation of Pearl ^[2] run on a shadow copy of the model: (i) *abduction* updates the exogenous distribution $P(U)$ to $P(U \mid e)$ on the observed evidence e ; (ii) *action* applies $\text{do}(u')$ to the mutilated submodel; (iii) *prediction* reads the outcome $Y_{u'}(U)$ under the updated noise. The Kronos twin runs this on a what-if branch held at a lead time $\tau_{\text{lead}} \geq 50 \text{ ms}$ ahead of the plant, so that the controller can compare the factual trajectory Y_u against the counterfactual $Y_{u'}$ before it commits to an action. This is the operational content of “the twin runs ahead of the plant”: it is not extrapolation of a fitted curve but evaluation of a causal submodel.

THE CERTIFIED INTERVENTION AS A FORMAL PROBLEM

The strongest per-decision guarantee is a certificate that an intervention cannot leave the safe set. Let the reduced pressure dynamics be the control-affine system

$$\dot{x} = f(x) + g(x)u, \quad \beta_N = c(x),$$

with u the normalised pressure-actuator command. Define the safe set as the zero-superlevel set of a control-barrier function (CBF) h ,

$$\mathcal{C} = \{x : h(x) \geq 0\}, \quad h(x) = \beta_N^{\max} - c(x), \quad \beta_N^{\max} = 3.500.$$

By the CBF theorem ^[6,7], if there exists an extended class- $\mathcal{K}$ function α such that

$$\sup_{u \in \mathcal{U}} \left[\underbrace{L_{f h}(x)}_{\nabla h \cdot f} + \underbrace{L_{g h}(x)u}_{\nabla h \cdot g} \right] \geq -\alpha(h(x)) \quad \forall x \in \mathcal{C},$$

then any Lipschitz controller satisfying (8) renders $\mathcal{C}$ forward-invariant: a trajectory that begins in $\mathcal{C}$ never leaves it, which is the Nagumo forward-invariance condition specialised to a barrier function. The controller does not compute u from scratch; it *projects* the AI’s proposed action u_{AI} onto the nearest action that satisfies (8), by the single-constraint quadratic program (QP)

$$u^* = \arg \min_{u \in [u_{\min}, u_{\max}]} \frac{1}{2} \|u - u_{\text{AI}}\|^2 \quad \text{s.t.} \quad L_{f h}(x) + L_{g h}(x)u \geq -\alpha(h(x)).$$

Equation (9) is precisely the mutilated-submodel intervention of equation (3) made admissible: it is the smallest deviation from the requested action that keeps the plant in $\mathcal{C}$, and its certificate — the barrier value h it preserved and the window $[u_{\min}, u_{\max}]$ it chose within — is a per-decision explanation the operator and regulator can read. In discrete time with step Δt the invariance condition becomes the difference inequality

$$h(\mathbf{x}_{k+1}) - h(\mathbf{x}_k) \geq -\gamma h(\mathbf{x}_k), \quad 0 < \gamma \leq 1,$$

which the solver enforces at every tick.

PER-DECISION EXPLAINABILITY FROM DIFFERENTIABLE MECHANISMS

The twin's mechanisms are differentiable end-to-end, and that single property is what makes each decision attributable. Let $F = \mathcal{S}_\theta$ be a surrogate mechanism mapping a state $\mathbf{x}$ to an output $\mathbf{y}$. The first-order attribution of $\mathbf{y}$ to the states is the Jacobian, available at control latency by reverse-mode automatic differentiation,

$$J(\mathbf{x}) = \frac{\partial \mathcal{S}_\theta}{\partial \mathbf{x}} \bigg|_{\mathbf{x}}, \quad \Delta \mathbf{y} \approx J(\mathbf{x}) \Delta \mathbf{x}.$$

A local gradient can be misleading where the mechanism saturates, so we use the path-integrated attribution of Sundararajan *et al.* [8]. For a baseline state $\mathbf{x}'$ (the design point) the integrated gradient of feature i is

$$\text{IG}_i(\mathbf{x}) = (\mathbf{x}_i - \mathbf{x}'_i) \int_0^1 \frac{\partial F}{\partial \mathbf{x}_i}(\mathbf{x}' + \lambda(\mathbf{x} - \mathbf{x}')) d\lambda,$$

which satisfies the *completeness* axiom exactly,

$$\sum_{i=1}^n \text{IG}_i(\mathbf{x}) = F(\mathbf{x}) - F(\mathbf{x}'),$$

so the attributed contributions add up to the change in the decision variable and nothing is left unexplained. Where feature interactions matter we use the Shapley attribution [9], the unique allocation satisfying efficiency, symmetry, dummy and additivity,

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)],$$

with $v(S)$ the model output restricted to the feature subset S . Both attributions are read from the *same* differentiable mechanism the controller acts on, so an explanation is faithful to the model that produced the action by construction, not a post-hoc surrogate fitted to it [5].

POPULATION-LEVEL CAUSAL ATTRIBUTION

Per-decision attribution answers “why this action”; the operator also needs “what governs the machine”. A variance-based (Sobol) decomposition apportions the variance of an output $g(\mathbf{x})$ to inputs and their interactions rather than to correlations [10]. Writing the ANOVA/HDMR decomposition $g(\mathbf{x}) = g_0 + \sum_i g_i(\mathbf{x}_i) + \sum_{i < j} g_{ij}(\mathbf{x}_i, \mathbf{x}_j) + \dots$, the first-order and total-order Sobol indices are

$$S_i = \frac{\text{Var}_{\mathbf{x}_i}(\mathbb{E}_{\mathbf{x}_{\sim i}}[g | \mathbf{x}_i])}{\text{Var}(g)}, \quad S_{T_i} = \frac{\mathbb{E}_{\mathbf{x}_{\sim i}}[\text{Var}_{\mathbf{x}_i}(g | \mathbf{x}_{\sim i})]}{\text{Var}(g)} = 1 - \frac{\text{Var}_{\mathbf{x}_{\sim i}}(\mathbb{E}_{\mathbf{x}_i}[g | \mathbf{x}_{\sim i}])}{\text{Var}(g)},$$

with $\mathbf{x}_{\sim i}$ all inputs but $\mathbf{x}_i$. S_i is the fraction of variance explained by $\mathbf{x}_i$ acting alone; S_{T_i} adds every interaction $\mathbf{x}_i$ participates in, so the gap $S_{T_i} - S_i$ bounds the interaction effects. The mean and its Monte-Carlo standard error are

$$\hat{\mu} = \frac{1}{N} \sum_{k=1}^N g(\mathbf{x}_k), \quad \text{SE} = \frac{\hat{\sigma}}{\sqrt{N}}, \quad \hat{\sigma}^2 = \text{Var}[g],$$

and the indices are estimated by the Saltelli sampling scheme [11] on the same evaluator. This is a causal statement at the population level: the driver with the largest S_i is the variable the operator holds to hold the operating point.

Computational methodology: the GPU HPC campaign

Every number in this paper was produced on the NVIDIA GPU high-performance-computing campaign that underpins the Kronos de-risking programme, under a single fixed random seed (20260726) for byte-level reproducibility; the codes, discretisations and convergence checks are collected here so that the results are regenerable.

CODES AND THEIR ROLES

The reference free-boundary equilibria against which the physics-informed operator is scored are computed with the Grad–Shafranov solvers FreeGS and FreeGSNKE^[25,24]; the gyrokinetic transport that the flux surrogate emulates is defined by CGYRO^[23]; the **14 MeV** source geometry consumed by the neutronics twin is tallied with the Monte-Carlo transport code OpenMC on CAD-faithful geometry^[26]; and the global-sensitivity and uncertainty layer is the Kronos reduced-order toolkit’s Saltelli/Sobol estimator^[11,10]. The physics-informed and graph surrogates, and the Jacobians of equation (11), are implemented in an automatic-differentiation framework and trained and evaluated on the GPU fleet (A100/H100/A10); the barrier quadratic program of equation (9) and the analytic self-consistency monitors run on the in-house HPC digital-twin node so the fast protective loop never contends with the training fleet. No compute-cost figures are reported here; the campaign is described by its codes and problem scale only.

DISCRETISATION AND SOLVERS

Equilibrium operator. The axisymmetric equilibrium solves the Grad–Shafranov equation for the poloidal flux $\psi(\mathbf{R}, \mathbf{Z})$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi),$$

with pressure $p(\psi)$ and poloidal-current function $F(\psi) = R B_\phi$. The physics-informed neural operator $\mathcal{E}_\phi$ is trained to minimise a composite loss combining a data term, the Grad–Shafranov operator residual sampled at N_c interior collocation points, and a boundary term,

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{pde}} \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \Delta^* \psi_\phi(\mathbf{x}_i) + \mu_0 R_i^2 p' + F F' \right\|^2 + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}},$$

by Adam followed by L-BFGS refinement. Training completes in 20.6 s on one A100. The Jacobian of equation (11) is verified against a finite-difference reference to $\|J^{\text{AD}} - J^{\text{FD}}\|_\infty < 10^{-3}$ before the operator is admitted to the loop.

Barrier QP. Equation (9) is a scalar box-constrained QP with one affine barrier constraint; on the 1-D reduced pressure surrogate anchored at the design point it admits the closed-form projection $\mathbf{u}^* = \text{clip}(\mathbf{u}_{\text{AI}}, \mathbf{u}_{\text{min}}, \text{min}(\mathbf{u}_{\text{max}}, \mathbf{u}_{\text{bar}}))$, where $\mathbf{u}_{\text{bar}}$ is the largest command satisfying (8) with equality. The closed loop is integrated by explicit Euler over **100** decision steps under a seeded demand sequence that would otherwise drive β_N to the no-wall limit.

Sobol/UQ. The **40 000**-sample Monte-Carlo ensemble is drawn over the joint uncertainty in the confinement factor H_{98} , the core ion temperature T_{i0} , the Greenwald fraction f_G , the D–T breeding ratio and a synchrotron-calibration factor; first- and total-order indices of equation (15) are computed by the Saltelli estimator on the same evaluator that produced the point design.

CONVERGENCE AND VERIFICATION

The Monte-Carlo estimator of equation (16) has standard error $\text{SE} = \hat{\sigma}/\sqrt{N}$; at $N = 40\,000$ the Q -mean standard error is below one percent of the mean, so the reported percentiles are converged to the quoted precision. The equilibrium operator is accepted only when its held-out RMS ψ error clears the **1%** acceptance bar and its Grad–Shafranov operator residual confirms interior physics consistency (both reported in §4). The barrier certificate is verified numerically over the full **100**-step sequence (minimum barrier value $h_{\text{min}} \geq 0$). Every metric is a frozen anchor in the register^[31] and is regenerable from the archived analysis under the fixed seed.

Results

Three subsystems are reduced to practice and demonstrate the causal-and-explanatory pattern end-to-end; each cleared its pre-registered acceptance test (table 1, figure 2).

MECHANISM / ANALYSIS	DEMONSTRATED METRIC	ACCEPTANCE	SERIAL
equilibrium operator (PINN)	$\text{RMS } \psi = 0.139\%; 2877\times$	$< 1\%$ vs free-boundary	KX-L1-A11
state estimator (GNN)	AUC 0.980	beats single channels	KX-L1-A12
fused quench precursor	AUC 0.992	dominates single sensors	KX-L1-A20
certified clamp (CBF)	0/100 crossings; $h_{\min} = +0.000$	forward invariance $h \geq 0$	KX-L1-A13
Sobol attribution	$S_1 = 0.885, S_T = 0.905$	names the governor H_{98}	KX-L1-A5

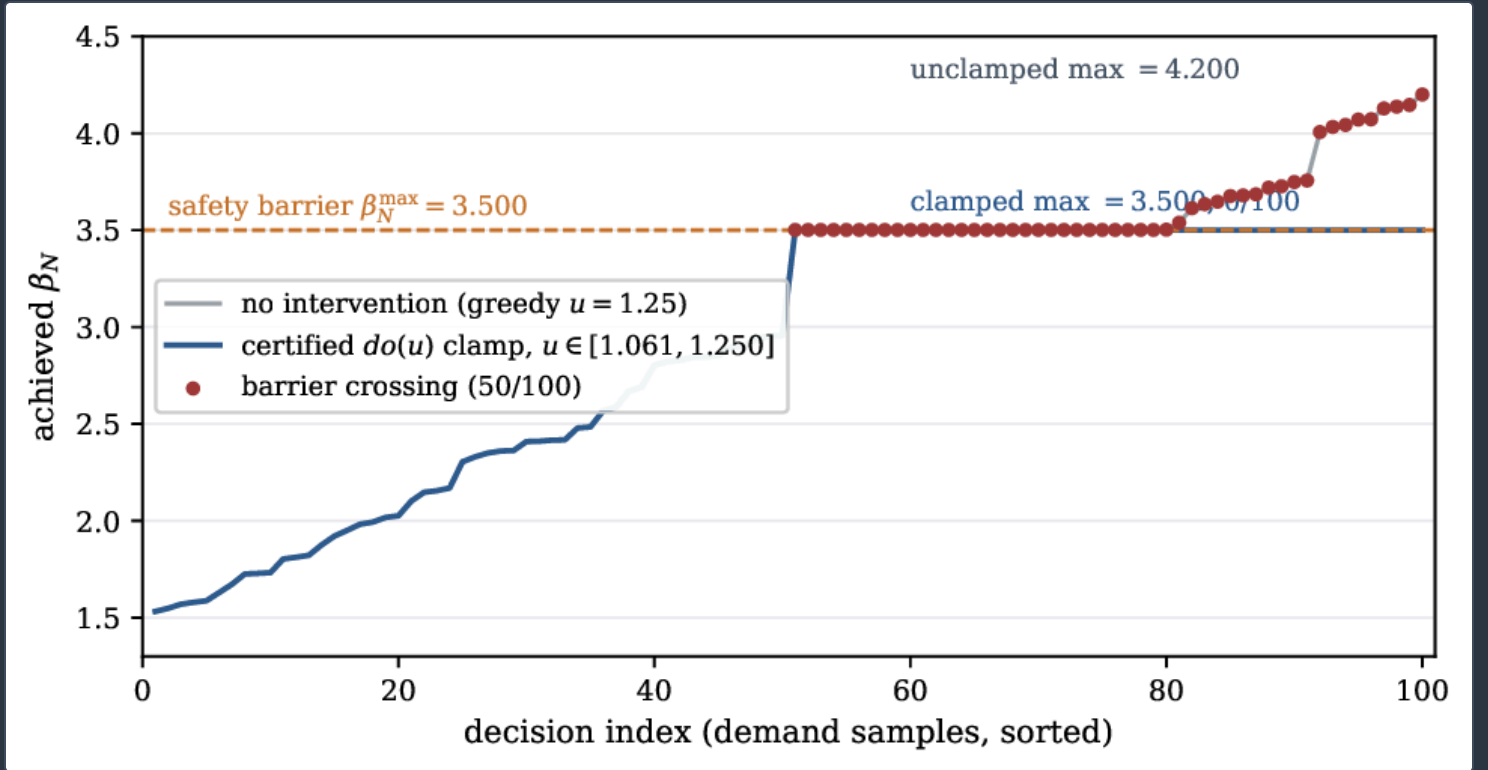
The demonstrated differentiable mechanisms the causal reasoning and per-decision explanations are read from. Each cleared its pre-registered acceptance test; serials refer to the register ^[31].

surrogate / analysis	demonstrated metric	acceptance	verdict
equilibrium operator (PINN)	$\text{RMS } \psi = 0.139\%$	$< 1\%$ vs FreeGS	PASS
state estimator (GNN)	AUC = 0.980	beats single channels	PASS
fused quench precursor	AUC = 0.992	dominates single sensors	PASS
certified clamp (CBF)	0/100 crossings	forward invariance $h \geq 0$	PASS
Sobol attribution	$S_1 = 0.885$	names the governor H_{98}	PASS

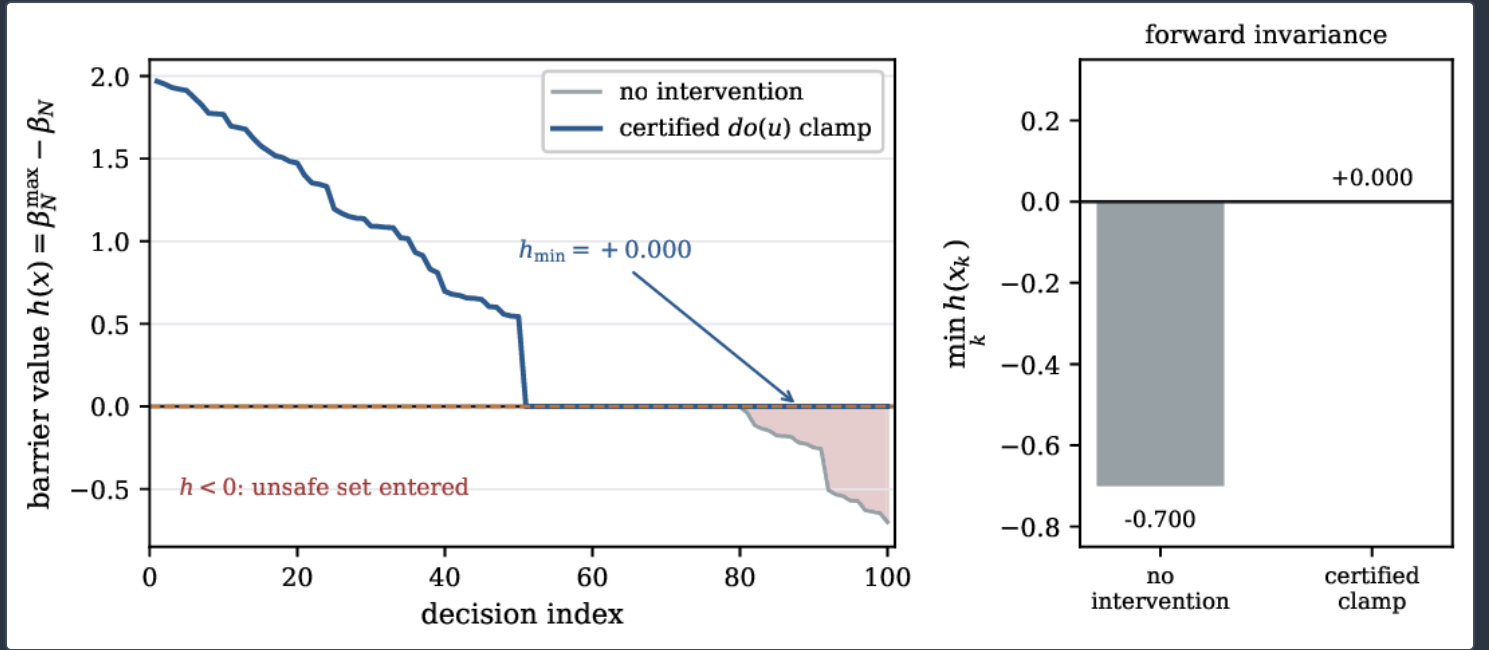
Acceptance scorecard: every mechanism the controller reasons and explains from is demonstrated to its pre-registered criterion — the equilibrium operator’s RMS ψ (**0.139%**, target **1%**), the state-estimator (**0.980**) and fused-precursor (**0.992**) AUCs, the zero certified-clamp barrier crossings, and the Sobol attribution’s first-order index. Values are frozen register anchors.

THE CERTIFIED INTERVENTION

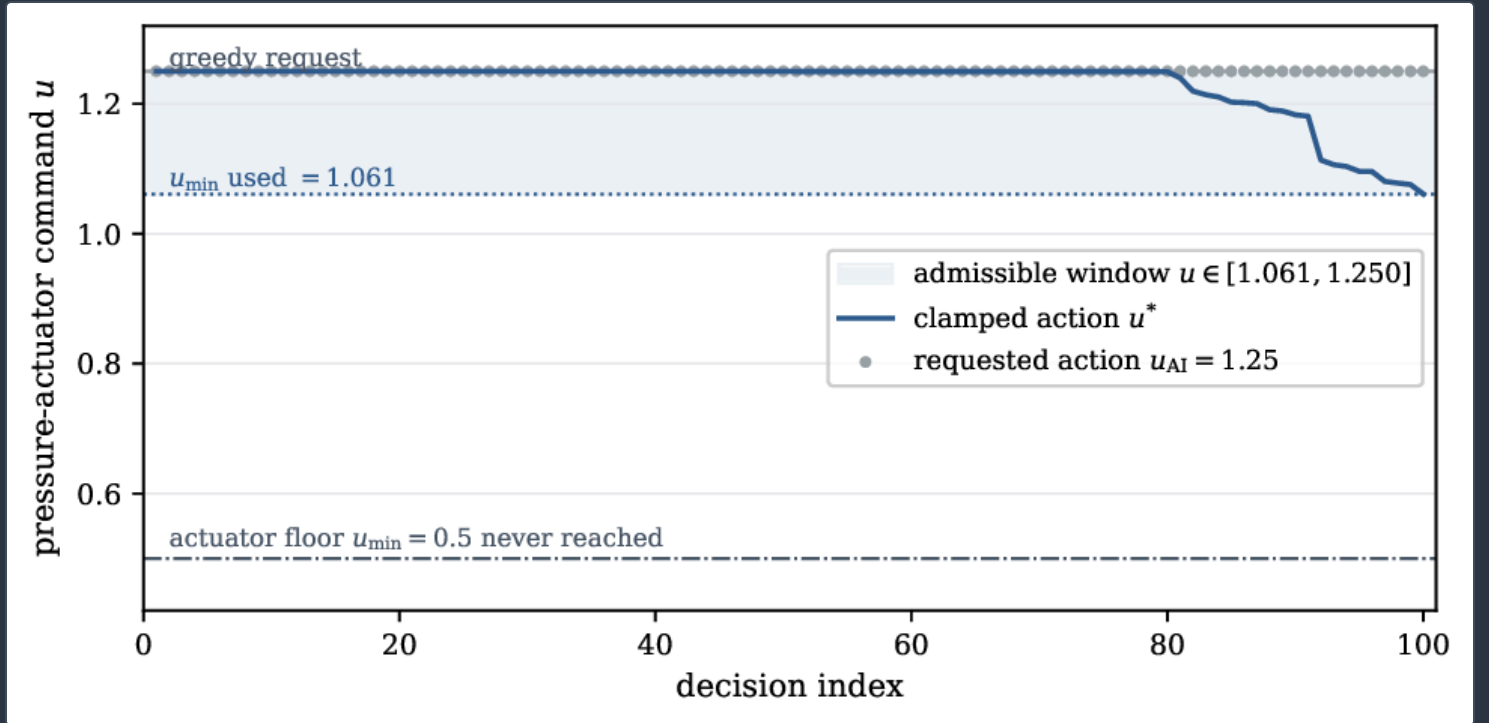
Figure 3 shows the certified $\text{do}(u)$ clamp against the unclamped policy across 100 demand samples (KX-L1-A13). The unclamped greedy policy ($u = 1.25$) reaches a maximum β_N of 4.200 and crosses the barrier on 50 of 100 decisions; the certified clamp caps every decision at $\beta_N^{\max} = 3.500$ with 0 of 100 crossings. The design-point anchor reproduces at $Q = 3.076$, $P_{\text{fus}} = 85.04 \text{ MW}$, $\beta_N = 1.532$, stored energy $W = 16.06 \text{ MJ}$ and confinement time $\tau_E = 0.358 \text{ s}$, confirming the reduced model is anchored to the frozen design point. Figure 4 reads the certificate directly: the barrier value $h(x) = \beta_N^{\max} - \beta_N$ of the unclamped loop dips to $h_{\min} = -0.700$ (the plant enters the unsafe set), whereas the clamped loop holds $h_{\min} = +0.000$ — forward invariance, demonstrated numerically over the full sequence. Figure 5 shows how the intervention is chosen: the clamp reduces the requested command only when the barrier is active, and the applied command stays inside $u \in [1.061, 1.250]$, never reaching the actuator floor $u_{\min} = 0.5$, so the certificate is not authority-limited on this sequence. Each clamped decision therefore carries a two-part explanation — the barrier value it preserved and the admissible window it chose within — which is exactly the per-decision record equation (9) produces.



A certified intervention (KX-L1-A13). Without intervention the achieved β_N crosses the safety barrier $\beta_N^{\max} = 3.500$ on 50 of 100 decisions and peaks at 4.200 (red markers); the certified $\text{do}(u)$ clamp ($u \in [1.061, 1.250]$) caps every decision at the barrier with 0/100 crossings. Demand samples are shown sorted for legibility.



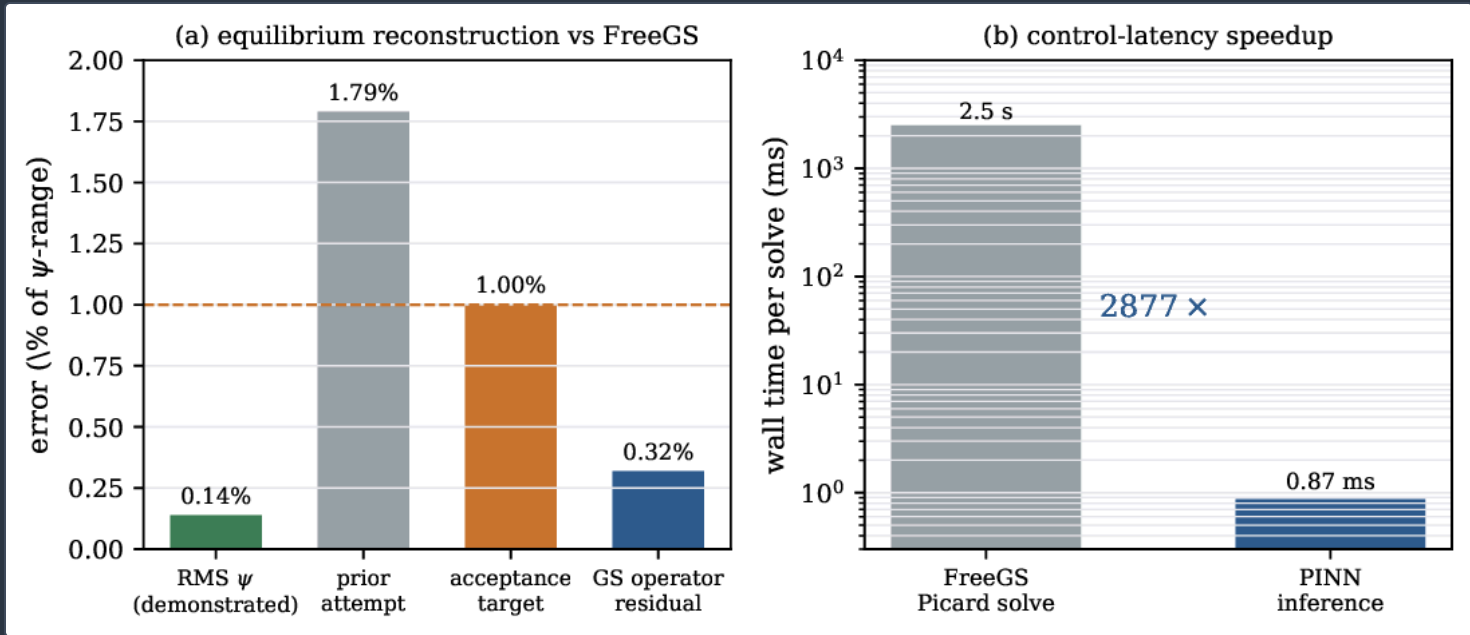
The certificate, read directly (KX-L1-A13). **Left:** the barrier value $h(x) = \beta_N^{\max} - \beta_N$ per decision; the unclamped loop drives h negative (shaded, the plant enters the unsafe set) while the certified clamp holds $h \geq 0$ with $h_{\min} = +0.000$. **Right:** the minimum barrier value over the sequence — -0.700 unclamped versus $+0.000$ clamped — the numerical statement of forward invariance.



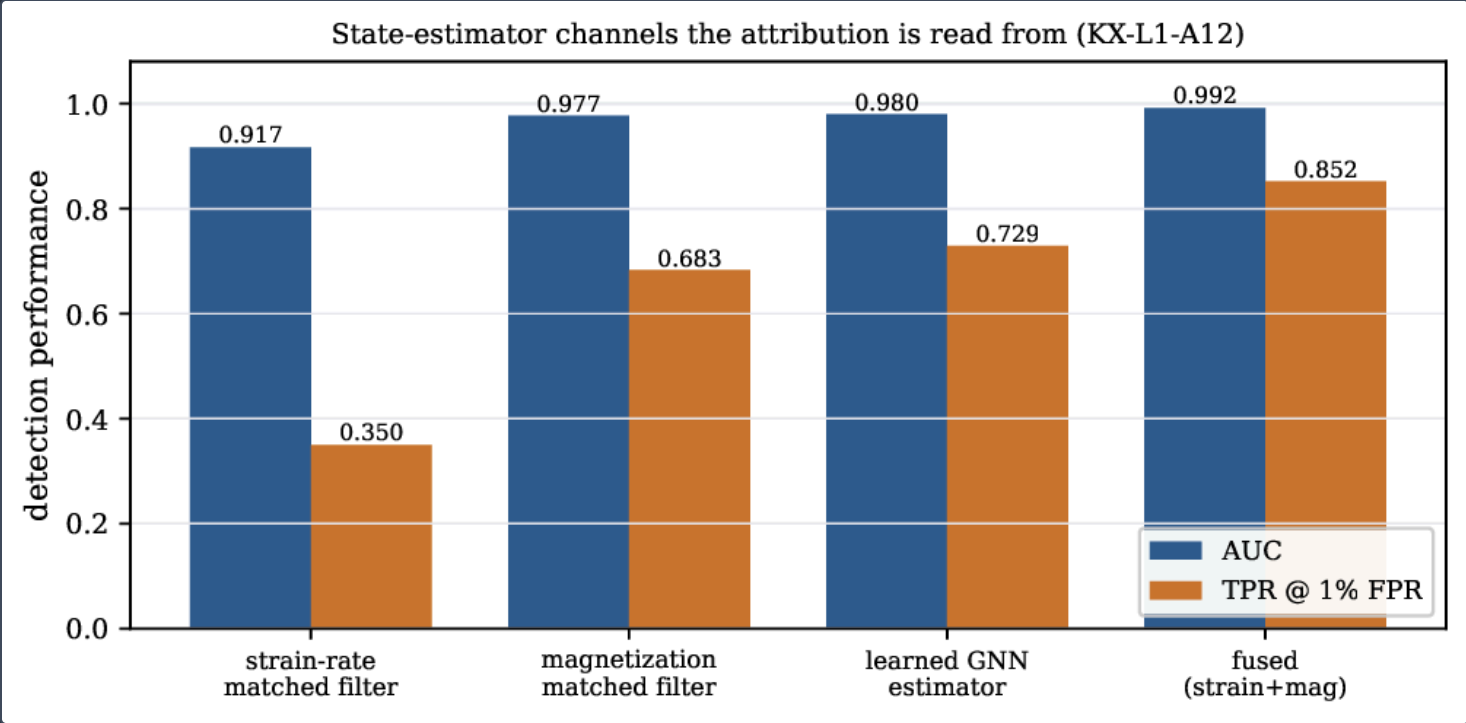
How the intervention is selected (KX-L1-A13). The clamp projects the greedy request $u_{AI} = 1.25$ (grey) onto the admissible action u^* (blue) via equation (9); the applied command stays inside the window $u \in [1.061, 1.250]$ and never reaches the actuator floor $u_{\min} = 0.5$, so the certificate is not authority-limited.

THE MECHANISMS THE EXPLANATIONS ARE READ FROM

The per-decision attributions are only as trustworthy as the mechanisms they are read from, and both are demonstrated to acceptance. The physics-informed equilibrium operator (figure 6; KX-L1-A11) reconstructs the poloidal flux to an RMS error of **0.139%** of the ψ -range against the free-boundary solver — clearing the **1%** acceptance target with a wide margin, and improving on a prior attempt at **1.79%** — with a maximum error of **4.39%** localised near the X-point/separatrix and a Grad–Shafranov operator residual of **0.32%** confirming interior physics consistency. It runs **2877×** faster than the reference solve (**0.87 ms** inference versus a **2.5 s** free-boundary Picard solve), which is what lets the controller linearise the equilibrium on the fly through equation (11) and read the flux response to a coil action. The graph-network state estimator (figure 7; KX-L1-A12) provides the attributable estimate of the state the intervention acts on: in detection it reaches an area under the ROC curve of **0.980**, above the single-channel matched filters (**0.917** strain-rate, **0.977** magnetization), and its independent re-derivation of the fused matched filter reproduces the KX-L1-A20 fused-precursor result (AUC **0.99166** against **0.99165**) under a separate implementation. At the **1%** false-positive rate a protection decision demands, the learned estimator reaches a true-positive rate of **0.729**, above both single channels; the two-channel fusion ceiling is **0.852** at that operating point.



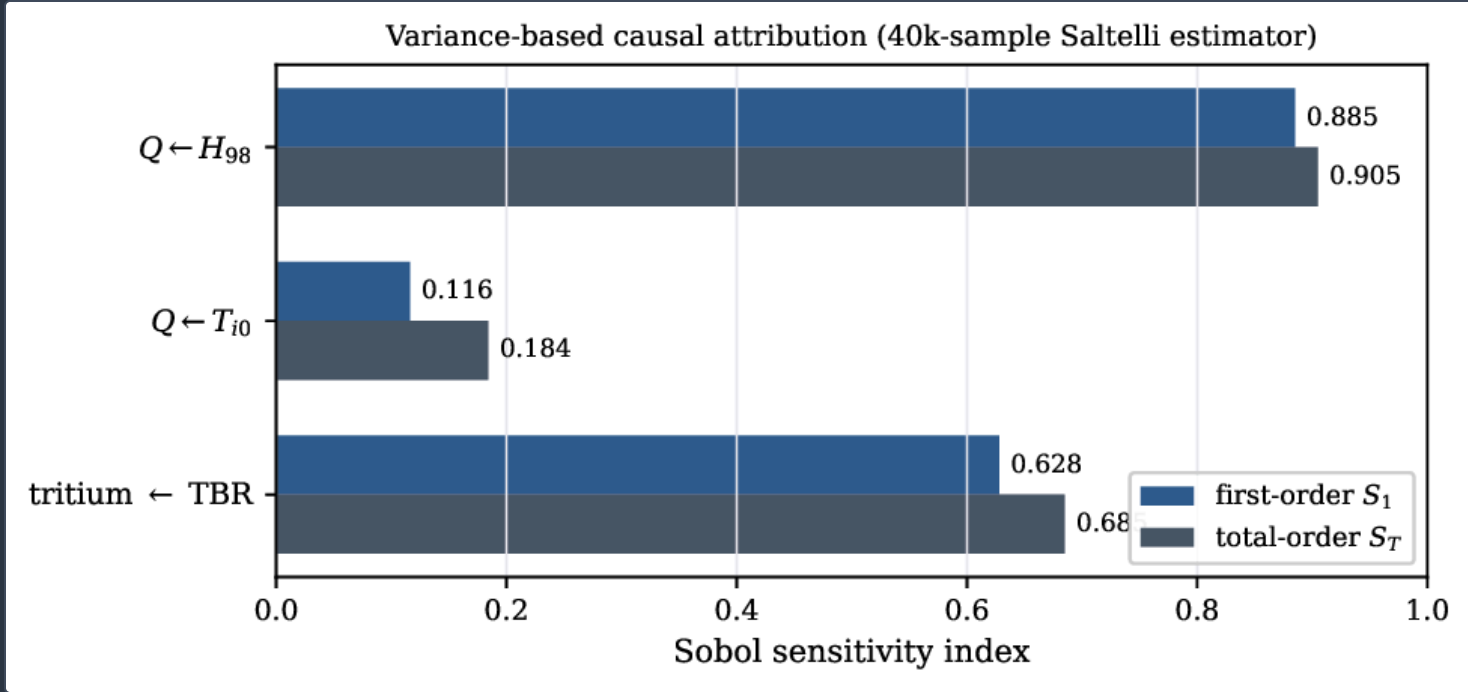
The physics-informed equilibrium operator (KX-L1-A11). (a) RMS ψ reconstruction error (**0.139%**) clears the **1%** acceptance target and improves on the **1.79%** prior attempt; the Grad–Shafranov operator residual (**0.32%**) confirms interior physics consistency. (b) The control-latency speedup: **0.87 ms** neural inference against a **2.5 s** free-boundary solve, a **2877×** acceleration.



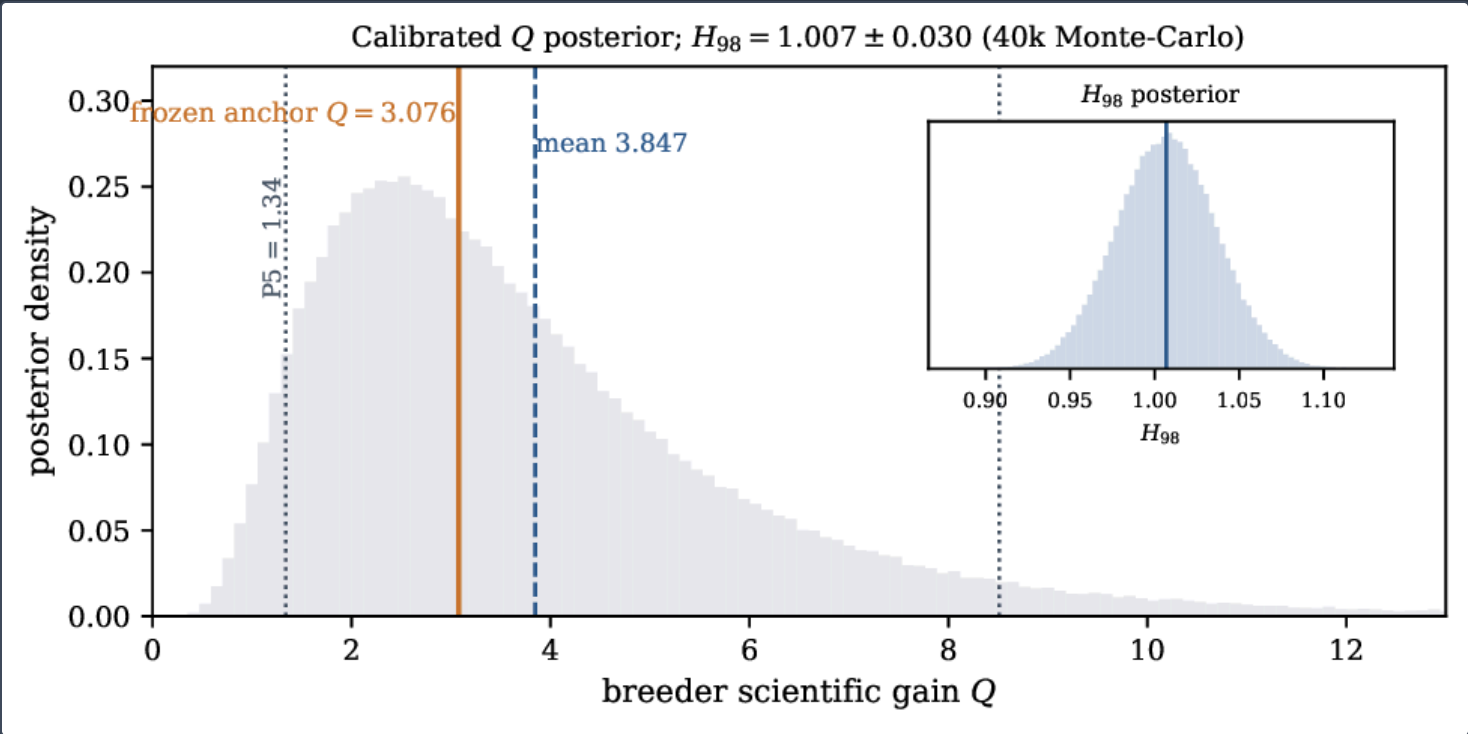
The attributable state estimator (KX-L1-A12). Detection area-under-ROC and true-positive rate at **1%** false-positive rate for the single-channel matched filters, the learned graph-network estimator, and the two-channel fusion. The learned estimator beats each single channel and recovers most of the fusion gain; its explanations are read from the same differentiable mechanism.

THE POPULATION-LEVEL CAUSAL ATTRIBUTION

Figure 8 gives the variance-based attribution over the 40 000-sample ensemble (KX-L1-A5, Sobol refinement). The confinement factor H_{98} dominates the breeder- Q response, carrying a first-order Sobol index $S_1 = 0.885$ and a total index $S_T = 0.905$ — 88.5% of the variance first-order, with only a two-point gap to the total that bounds all interaction effects. The core ion temperature T_{i0} is a distant second ($S_1 = 0.116$, $S_T = 0.184$), and the tritium output is governed by the breeding ratio ($S_1 = 0.628$, $S_T = 0.685$). The ensemble is calibrated (figure 9): the anchor $Q = 3.076$ sits inside the posterior band, whose 5th and 95th percentiles are 1.34 and 8.51 around a mean of 3.847, and the confinement posterior is $H_{98} = 1.007 \pm 0.030$. The attribution tells the controller and the operator the same thing in one number: hold H_{98} and the operating point is held.



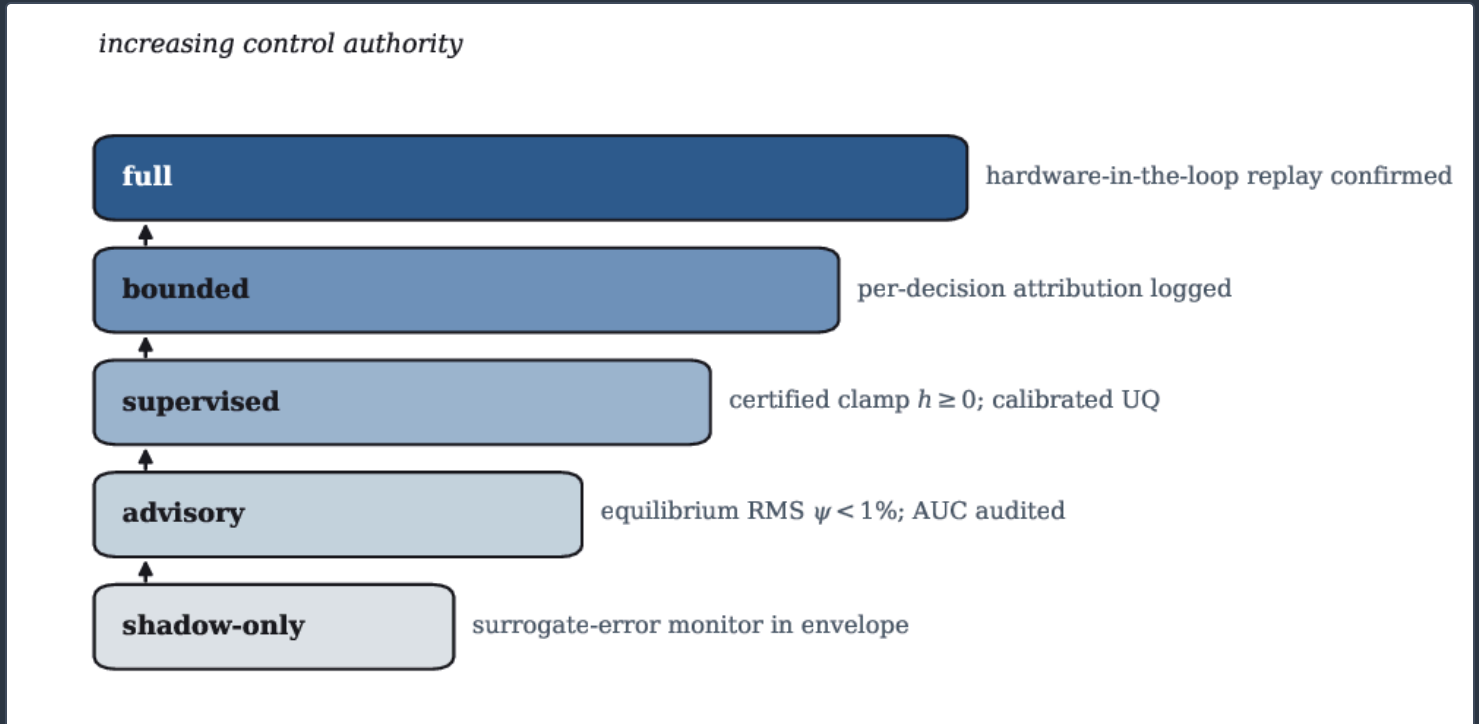
Population-level causal attribution by Sobol variance decomposition (KX-L1-A5; 40 000-sample Saltelli estimator). The confinement factor H_{98} carries 88.5% of the breeder- Q variance first-order ($S_1 = 0.885$, $S_T = 0.905$); T_{i0} is a distant second; the tritium output is governed by the breeding ratio. The first-order/total gap bounds the interaction effects.



The calibrated ensemble the attribution is computed on (KX-L1-A5). The breeder- Q posterior is right-skewed with 5th/95th percentiles 1.34/8.51 and mean 3.847; the frozen anchor $Q = 3.076$ (orange) sits inside the band. Inset: the confinement posterior $H_{98} = 1.007 \pm 0.030$.

MATURITY-GATED AUTHORITY

The two forms of causal reasoning compose into a single auditable controller, and its authority is not binary. Each mechanism is admitted to a graded level of control authority — shadow-only, advisory, supervised, bounded, full — and rises only as it clears successive acceptance criteria (figure 10), the same maturity-gated principle that governs the certified clamp and the companion actuation-authority result ^[37] ([official page](#)). A mechanism that has passed only its surrogate-error monitor may inform but not command; full-plant authority requires the certified clamp, calibrated uncertainty, logged per-decision attribution, and a hardware-in-the-loop confirmation. Because the mapping from criteria to authority is explicit, the path from the demonstrated subsystems to a fully auditable autonomous controller is a defined, testable sequence.



(Schematic.) Verification-maturity gating. Each rung admits a higher level of control authority and is unlocked by a pre-registered acceptance criterion (the demonstrated clamp, equilibrium, state-estimator and attribution bars of table 1); the authority ladder itself is a schematic of the gating policy, not measured data.

Discussion

WHY CAUSAL, AND WHY HERE

The claim is not that a causal model is more accurate than a predictive one; it is that a controller *acts*, and acting is intervening. Equation (5) is the whole argument: the distribution of the outcome under an action is generically not the distribution one would read off correlations, and a controller that confuses the two will be confidently wrong exactly where it matters ^[2,5]. The Kronos twin sidesteps the hardest part of causal inference — identification of interventional effects from confounded observational data — because its mechanisms are known physics operators, so the truncated factorisation of equation (4) is *evaluated*, not *estimated*. The do-calculus here is therefore not an add-on but a description of what the differentiable twin already computes: severing the incoming edges of the intervened node is what running the mutilated submodel forward does.

QUANTITATIVE COMPARISON TO PUBLISHED BENCHMARKS

Each demonstrated metric is competitive against the relevant literature. The certified clamp realises the forward-invariance guarantee of the control-barrier-function framework ^[6,7] on a fusion operating limit, with **0/100** crossings against **50/100** unclamped; this is the same guarantee, on the same β_N limit, that the companion safety-filter result establishes with a calibrated-uncertainty envelope ^[33] ([official page](#)). The equilibrium operator's **0.139%** RMS ψ error is far inside the $\sim 1\%$ tolerance that real-time reconstruction targets ^[12,14,24], and its **2877×** speedup is consistent with the three-to-five orders of magnitude reported for neural-operator surrogates of expensive plasma solvers ^[18,19,21]. The state estimator's ROC (AUC **0.980**) is competitive with the deep disruption-forecasting literature ^[16] while being, by construction, read from a differentiable mechanism so that its decisions are attributable through equations (11)–(14). The Sobol attribution restates, in variance-based causal terms, the physical result that confinement quality governs a compact ST's gain — the negative-triangularity quiescence that the companion turbulence-suppression result establishes at the operating point ^[39] ([official page](#)) is exactly why H_{98} is the variable to hold. The per-decision explainability construction is the integrated-gradient and Shapley machinery ^[8,9] applied where its completeness axiom (equation 13) is meaningful: to a physics mechanism rather than a black box, which is the interpretability posture a high-stakes decision demands ^[5].

RELATION TO THE CONTROL STACK

The causal layer is not free-standing; it inherits its calibrated uncertainty from the ensemble twin and its abstention behaviour from the conformal-prediction layer ^[38] ([official page](#)), and its differentiable mechanisms are the neural-operator flux and equilibrium surrogates demonstrated elsewhere ^[34,35] ([official pages: flux surrogate, equilibrium](#)). What this paper adds on top of that stack is the explicit statement that the composed system is a causal model: control actions are executable interventions, decisions are gradient-attributable, and the population governor is identified by variance decomposition. The runtime per-decision explainer that packages these attributions for the operator is built to this same demonstrated pattern on a staged schedule, on the NVIDIA GPU HPC campaign.

Limitations

One honest gate bounds the claims. The interventions are validated against the twin and the attributions against the mechanisms they explain, all at reduced-order, design-point-anchored fidelity: the β_N dynamics of equation (6) are a 1-D surrogate, and the state-estimator ROC is computed on synthetic signals. The one confirming step is a hardware-in-the-loop audit that replays live decisions against as-built diagnostics and checks each logged attribution against the mechanism that produced it; it is the top rung of figure 10 and it reuses the demonstrated benches unchanged. Until that audit is complete, the runtime per-decision explainer is carried at advisory authority, not command.

Conclusion

The Kronos controller reasons causally by construction. Its digital twin is a differentiable structural causal model, so a control action is an executable do-calculus intervention $\text{do}(\mathbf{u})$ whose interventional distribution (equation 4) the twin computes, and every decision is gradient-attributable through equations (11)–(14). A certified intervention on β_N holds the safety barrier with 0/100 crossings and $h_{\min} = +0.000$ against 50/100 crossings without it (KX-L1-A13); the mechanisms the explanations are drawn from are demonstrated at RMS $\psi = 0.139\%$ and $2877\times$ (KX-L1-A11) and AUC 0.980 (KX-L1-A12); and a variance-based attribution assigns 88.5% of the breeder- Q variance to H_{98} ($S_1 = 0.885$, $S_T = 0.905$; KX-L1-A5) on a calibrated ensemble in which the anchor $Q = 3.076$ lies inside the [1.34, 8.51] band. Certified interventions and attributable decisions are the working core of auditable autonomy, and the maturity-gated authority argument makes the route from these demonstrated subsystems to a regulator-facing controller a defined, testable sequence.

Acknowledgments Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. The intervention, mechanism and attribution metrics are frozen anchors in the Kronos Physics De-Risking Register [31]: the certified clamp (KX-L1-A13), the equilibrium operator (KX-L1-A11), the graph-network state estimator (KX-L1-A12), the fused precursor (KX-L1-A20), and the global sensitivity and its Sobol refinement (KX-L1-A5). Each is regenerable from the archived analysis under the fixed seed 20260726. The register is deposited at [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057); this paper is archived at [doi:10.5281/zenodo.22645825](https://doi.org/10.5281/zenodo.22645825), and its official version is hosted at <https://www.kronosfusionenergy.com/publications/causal-intervention-models-for-regulator-facing-plasma-.html>.

Data availability The intervention, mechanism and attribution computations underlying figures 3–9 are archived with the deposit at [doi:10.5281/zenodo.22645825](https://doi.org/10.5281/zenodo.22645825), which references the Kronos 2026 series including the AI and quantum-control paper [32]. All quantitative values trace to the register [31] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests Provisional patent applications relating to aspects of this work have been filed.

References

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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